

All Definition

01. Definition:

(i) Singular Value Decomposition :-

The singular value decomposition of a matrix A is the factorization of A into the product of three matrices $A = UDV^T$ where the column of U and V are orthogonal matrices and D is diagonal matrices with positive real entries.

(ii) Eigen Value Decomposition :-

In linear algebra, eigen decomposition or sometimes spectral decomposition is the factorization of a matrix, into a canonical form, whereby the matrix is represented in terms of its eigen values and eigen vectors. It can be written as, $A = PDP^{-1}$.

(iii) LU decomposition :-

If any linear system of equation can be performed as $AX = b$ without row interchanges, then A can be factored into the product of a lower triangular matrix L and an upper triangular matrix U . i.e. $A = LU$ which is known as LU decomposition or LU factorization.

(i) Vector :-

A vector is a quantity having both magnitude and direction. Such as displacement, velocity, force and acceleration etc.

(ii) Scalar :- A scalar is a quantity having magnitude but no direction. Such as mass, length, time, temperature and any real numbers.

(iii) Unit vector :-

A unit vector is a vector having unit magnitude. If $\vec{A}$ is a vector with magnitude $A \neq 0$, then $\vec{A}/A$ is a unit vector having the same direction as $\vec{A}$.

$$\text{Unit vector } \hat{a} = \frac{\vec{A}}{|\vec{A}|}$$

(iv) Dot product or (v) Scalar product :-

The Dot or scalar product of two vectors $\vec{A}$ and $\vec{B}$ denoted by $\vec{A} \cdot \vec{B}$ (read $\vec{A}$ dot $\vec{B}$) is defined as the product of magnitude of $\vec{A}$ and $\vec{B}$ and the cosine of the angle θ between them.

$$\text{In symbols, } \vec{A} \cdot \vec{B} = AB \cos \theta, \quad 0 \leq \theta \leq \pi$$

(vi) Vector field :-

If to each point (x, y, z) of a region R in space; there corresponds a vector $\vec{v}(x, y, z)$, then $\vec{v}$ is called a vector function and we say that a vector field $\vec{v}$ has been defined in R .

(vii) Scalar field :-

If to each point (x, y, z) of a region R in space there corresponds a number or scalar $\phi(x, y, z)$, then ϕ is called a scalar function of position or scalar point function and we say that a scalar field ϕ has been defined in R .

④ Projection (अनुप्रसर)

$$\text{Projection of } \vec{A} \text{ on } \vec{B} = \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|}$$

$$\text{Projection of } \vec{B} \text{ on } \vec{A} = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}|}$$

④ Component (अंश)

$$\text{Component of } \vec{A} \text{ on } \vec{B} = \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} \cdot \hat{b} \left[\hat{b} = \frac{\vec{B}}{|\vec{B}|} \right]$$

$$\text{Component of } \vec{B} \text{ on } \vec{A} = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}|} \cdot \hat{a} \left[\hat{a} = \frac{\vec{A}}{|\vec{A}|} \right]$$

Definition:-

④ Del/Nabla: The vector differential operator Del, written $\vec{\nabla}$, is defined by,

$$\vec{\nabla} \equiv \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

$$\equiv \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

The operator $\vec{\nabla}$ is also known as nabla.

④ The Gradient: Let $\phi(x, y, z)$ be defined and differentiable at each point (x, y, z) in a certain region of space. The gradient of ϕ , written $\vec{\nabla} \phi$ or $\text{grad } \phi$ is defined by,

$$\vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi$$

$$= \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$$

④ The Divergence: Let $\vec{v}(x, y, z) = v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}$ be defined and differentiable at each point (x, y, z) in a certain region of space. Then the divergence of $\vec{v}$, written $\vec{\nabla} \cdot \vec{v}$ or $\text{div } \vec{v}$, is

defined by,

$$\vec{\nabla} \cdot \vec{v} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k})$$

$$= \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

The Curl: If $\vec{v}(x, y, z)$ is a differentiable vector field then the curl or rotation of $\vec{v}$, written $\vec{\nabla} \times \vec{v}$, curl $\vec{v}$ or rot $\vec{v}$, is defined by,

$$\vec{\nabla} \times \vec{v} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}$$

$$= \hat{i} \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_2 & v_3 \end{vmatrix} - \hat{j} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ v_1 & v_3 \end{vmatrix} + \hat{k} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ v_1 & v_2 \end{vmatrix}$$

$$= \left(\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z} \right) \hat{i} - \left(\frac{\partial v_3}{\partial x} - \frac{\partial v_1}{\partial z} \right) \hat{j} + \left(\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y} \right) \hat{k}$$

Green's TheoremDefinition:

Let R be a simple connected region with a piecewise smooth boundary C , oriented counter clockwise. If $M(x, y)$ and $N(x, y)$ be continuous and have continuous partial derivatives, in an open region containing R , then:

$$\oint_C (M dx + N dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

Q. State Divergence theorem (Gauss Theorem)

Solution:

Statement: For a close surface, enclosing a region V in a vector field $\vec{F}$, then

$$\int_V \text{div } \vec{F} dv = \int_S \vec{F} \cdot d\vec{s} \text{ which is called the divergence theorem.}$$